

Question ID: 9bedc4a0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

A company that produces socks wants to estimate the percent of the socks produced in a typical week that are defective. A random sample of **310** socks produced in a certain week were inspected. Based on the sample, it is estimated that **12%** of all socks produced by the company in this week are defective, with an associated margin of error of **3.62%**. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all socks produced by the company during this week?

Answer

- A. **3.62%** of the socks are defective.
- B. It is plausible that between **8.38%** and **15.62%** of the socks are defective.
- C. **12%** of the socks are defective.
- D. It is plausible that more than **15.62%** of the socks are defective.

Correct Answer: B

Rationale

Choice B is correct. It’s given that, based on the sample, an estimate of **12%** of all socks produced by the company in a certain week are defective, with an associated margin of error of **3.62%**. This estimate, plus or minus the margin of error, gives an interval of plausible values for the actual percent of all socks produced by the company that week that are defective. Subtracting **3.62%** from **12%** yields **8.38%**. Adding **3.62%** to **12%** yields **15.62%**. Therefore, it is plausible that between **8.38%** and **15.62%** of all socks produced by the company are defective.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. **12%** is the estimated percent of defective socks based on the sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the percent of defective socks is exactly **12%** but instead that it lies in an interval of plausible percents.

Choice D is incorrect and may result from conceptual errors.